BIOINFORMATICS: FROM MULTI-OMICS DATA TO DISCOVERY

Week 12

Interpretation & Writing I
From Plots to Text

Reading Scientific Papers

Writing from Multi-Omics Evidence

Fall 2026

60-Minute Session

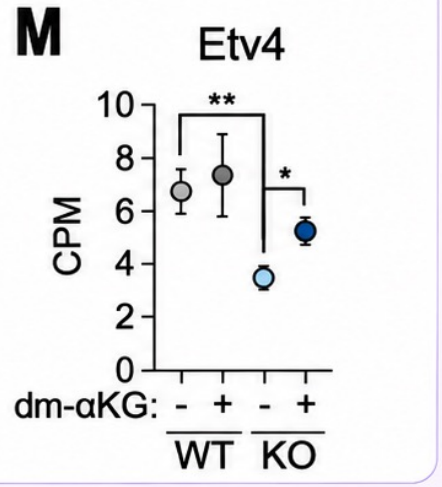
Dr. Liwei Xie



Every scientific claim should be traceable to its evidence.

From Reading to Writing

READ THE FIGURE



EVALUATE THE CLAIM

- Comparison
- Direction
- Uncertainty
- Scope

WRITE THE RESULT

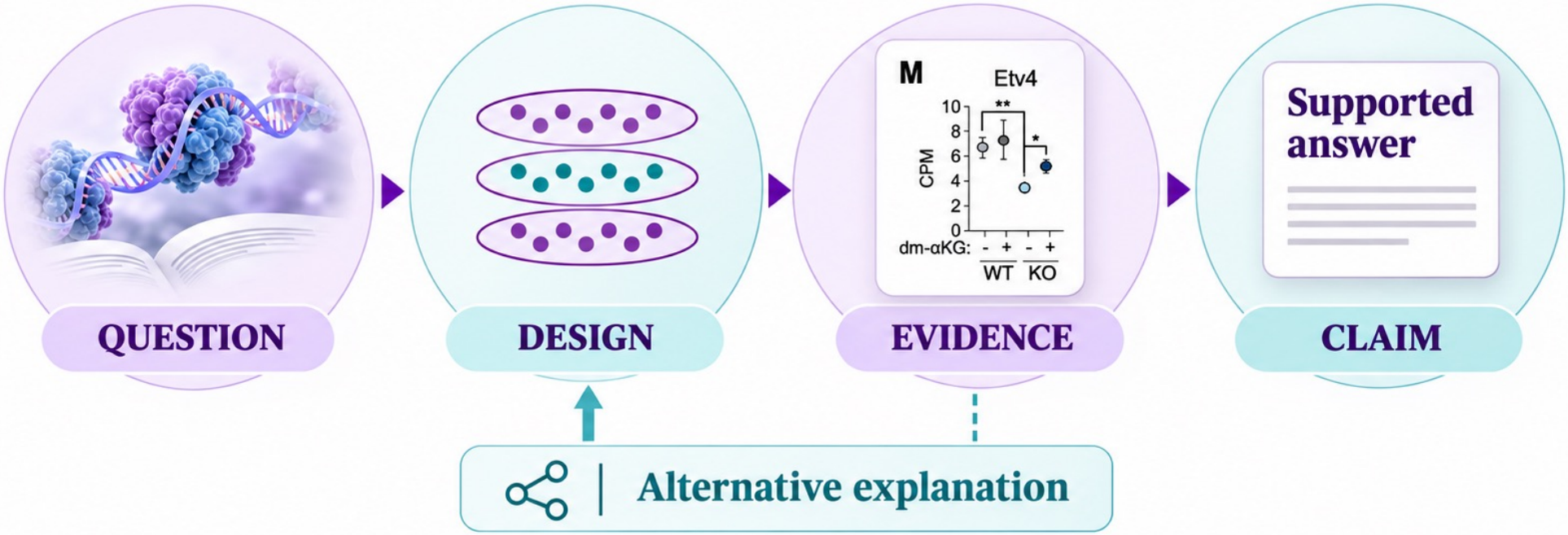
Observation
dm-αKG increased Etv4 expression in knockout brains.

Interpretation
Direct mediation remains unresolved.

Figure: Sternisha et al., Science (2026), Fig. 6M. Teaching annotations added.

Accurate writing begins with accurate reading of the data.

The Argument Behind a Paper



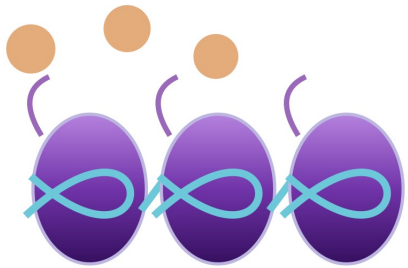
Scientific argument schematic; figure excerpt: Sternisha et al., Science (2026), Fig. 6M.

A paper connects a scientific question to a supported answer.

Question, Gap, and Hypothesis

Known

αKG



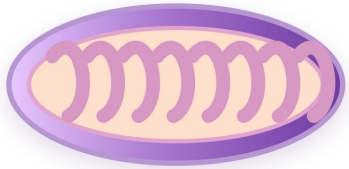
αKG supports demethylation

Gap

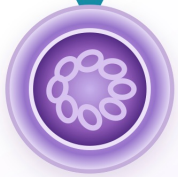


Nuclear αKG regulation

Hypothesis



Proposed



Mitochondrial αKG supply

Conceptual teaching schematic informed by Sternisha et al., Science (2026).

A useful hypothesis predicts a result that experiments can test.

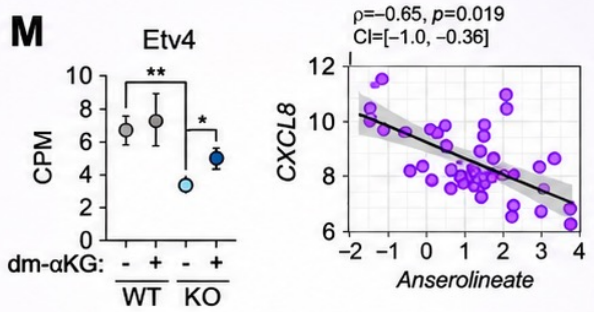
Three Passes Through a Paper

ORIENT

Question and main claim

What is tested?

TRACE



Figures and Results

AUDIT

Methods

Supplements

Limitations

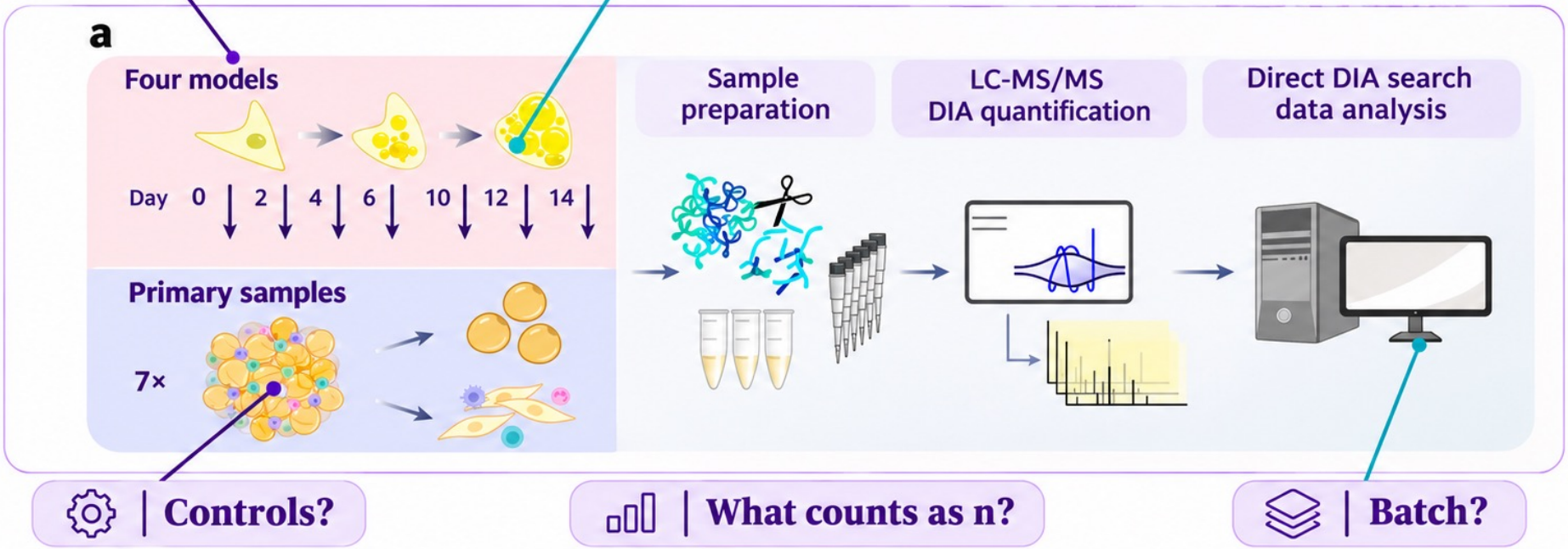
Course reading strategy; Carey et al., PLOS Computational Biology (2020). Figure excerpts: Sternisha (2026); Priya (2022).

Each reading pass should answer a different scientific question.

The Design Behind the Data

Biological unit?

Comparison?



Controls?

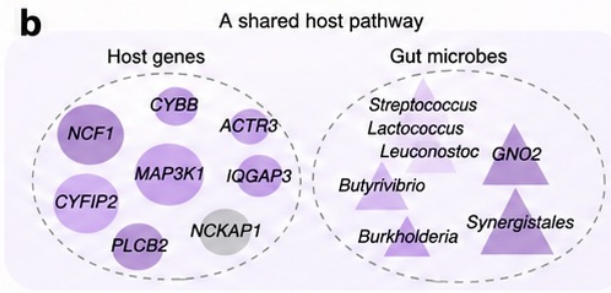
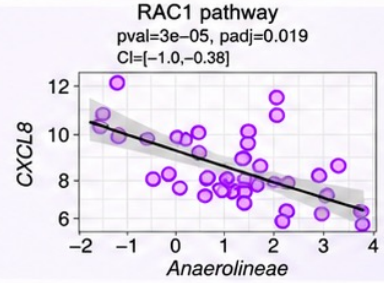
What counts as n?

Batch?

Klingelhuber et al., Nature Metabolism (2024), Fig. 1a. Four cell models; primary samples from seven donors.

The experimental unit determines what counts as independent evidence.

A Map of Claims and Evidence

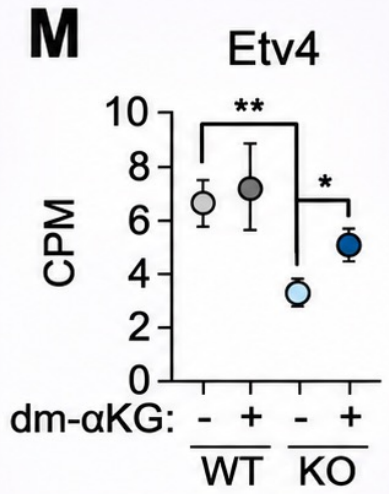
CLAIM	SUPPORTING PANEL	REMAINING QUESTION
Pathway association		**Shared pathway, same mechanism?**
Gene–taxon association		**Causal direction unresolved**

Priya et al., Nature Microbiology (2022), selected Fig. 2b cohort and Extended Data Fig. 2 association.

Link each claim to the panel that actually supports it.

Eight Questions for One Figure

- **Biological question?**
- **Measured variable?**
- **Comparison groups?**
- **Display and units?**



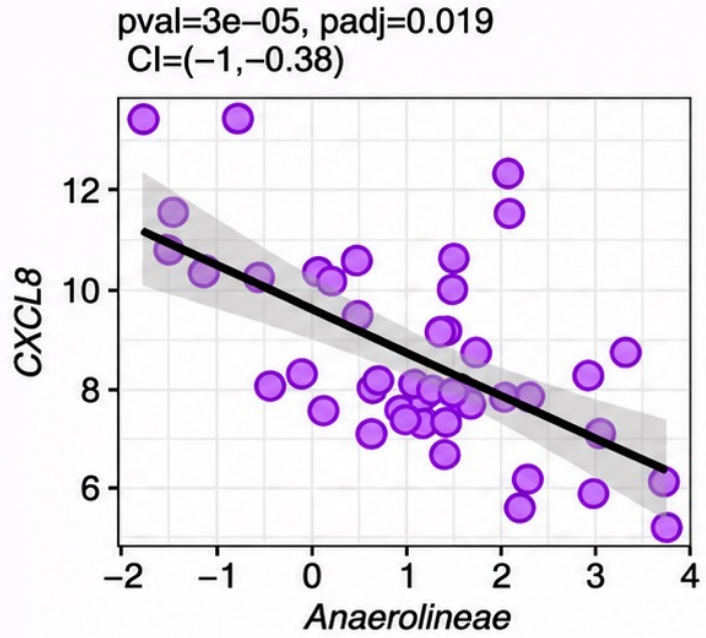
Mouse brain RNA-seq
Group means ± SEM

- **Statistical evidence?**
- **Interpretation?**
- **Remaining uncertainty?**
- **Next experiment?**

Sternisha et al., Science (2026), Fig. 6M. Vehicle: n = 10/group; dm-αKG: n = 8/group. Points are group means.

Separate what the figure shows from what you infer.

Effect Size, Uncertainty, and FDR



Effect direction
Negative association

Coefficient interval
95% CI: −1 to −0.38

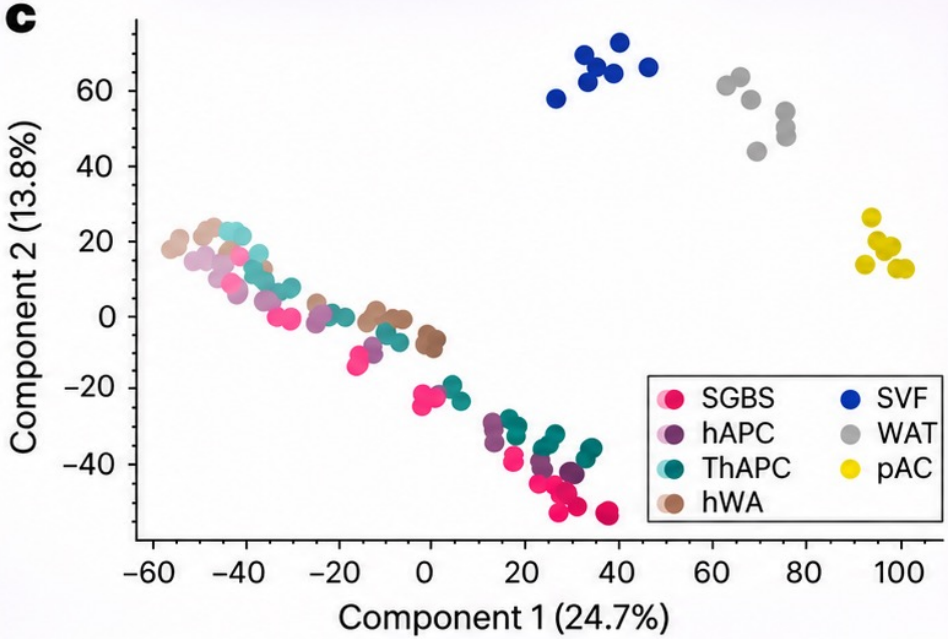
Multiple testing
BH-adjusted P = 0.019

Gray band
Fitted-line uncertainty

Priya et al., Nature Microbiology (2022), Extended Data Fig. 2. Numerical CI/P: desparsified lasso; gray band: linear fit.

Effect size, precision, and multiple testing answer different questions.

What Does This Plot Actually Show?



What is a point?
A sample

What defines distance?
Differences in PC scores

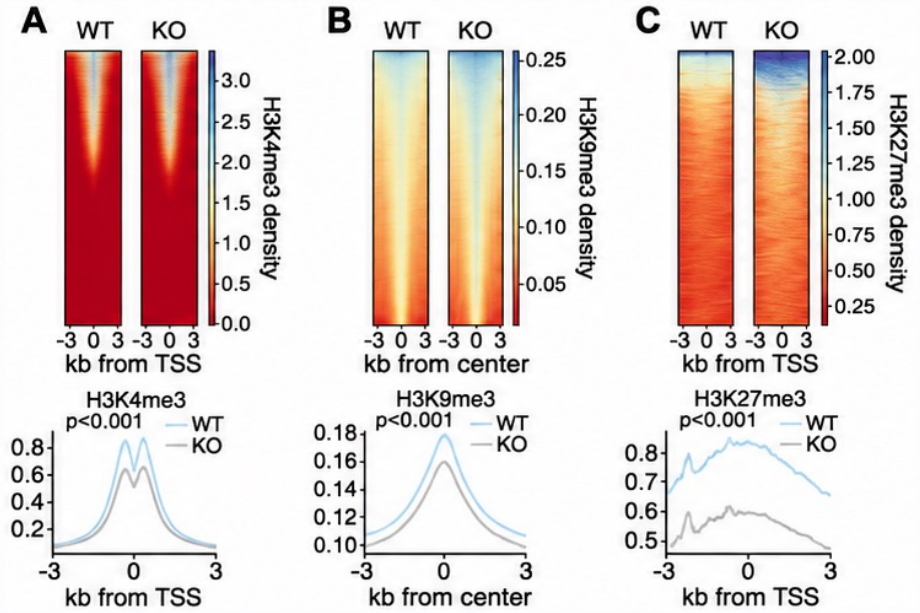
Why the separation?
Biology or batch?

Klingelhuber et al., Nature Metabolism (2024), Fig. 1c. Original axes, variance percentages, colors, and legend retained.

PCA displays variation that requires biological and technical interpretation.

Genomic and Regulatory Evidence

Histone-mark profiles



From signal to interpretation

Chromatin profile

↓

Differential signal?

↓

Candidate gene

↓

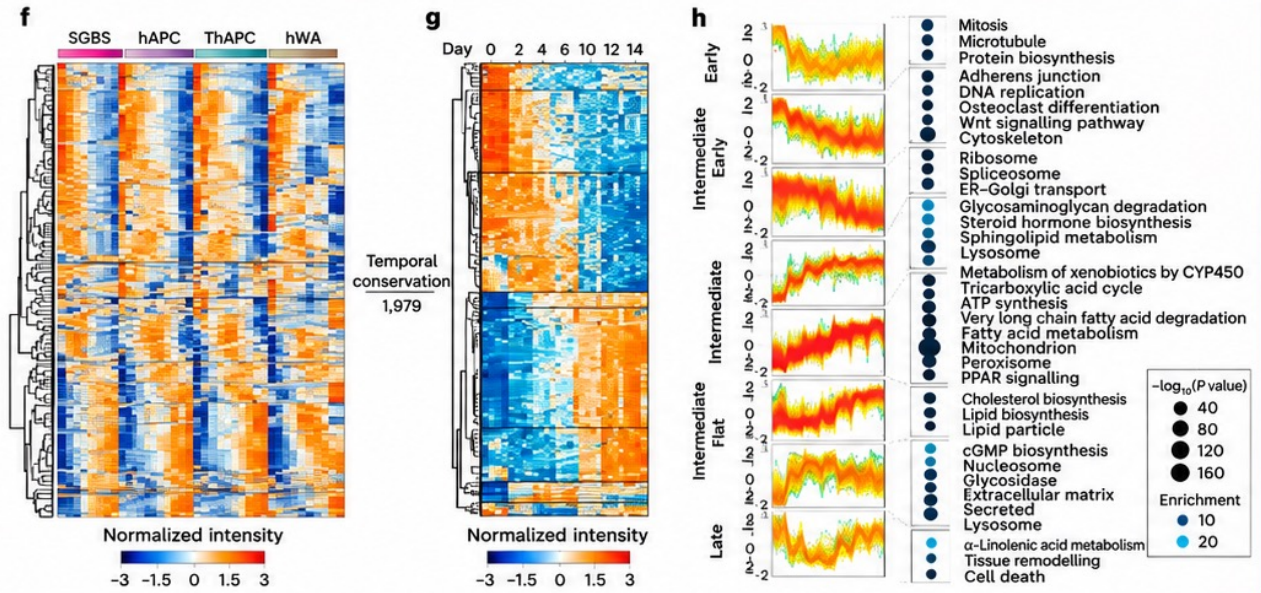
Functional test needed

Sternisha et al., Science (2026), Fig. 6A–C. Original chromatin profiles retained.

Genomic location prioritizes hypotheses; function needs testing.

RNA, Protein, and Metabolite Evidence

Temporal proteome: original figure



Interpretation boundaries

Transcript ≠ protein
Time lag and turnover

Abundance ≠ activity
Enzyme function is distinct

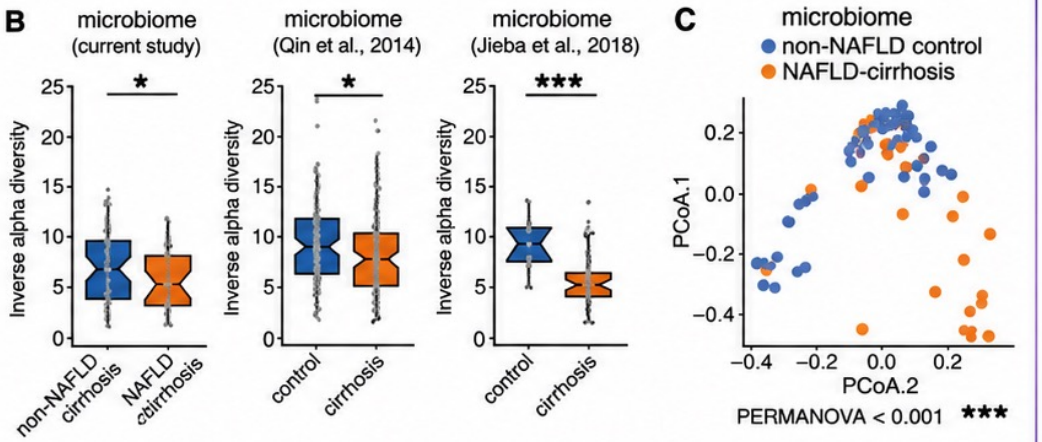
Pool ≠ flux
Metabolite turnover is unseen

Klingelhuber et al., Nature Metabolism (2024), Fig. 1f–h; RNA–protein time lag discussed in Extended Data Fig. 4.

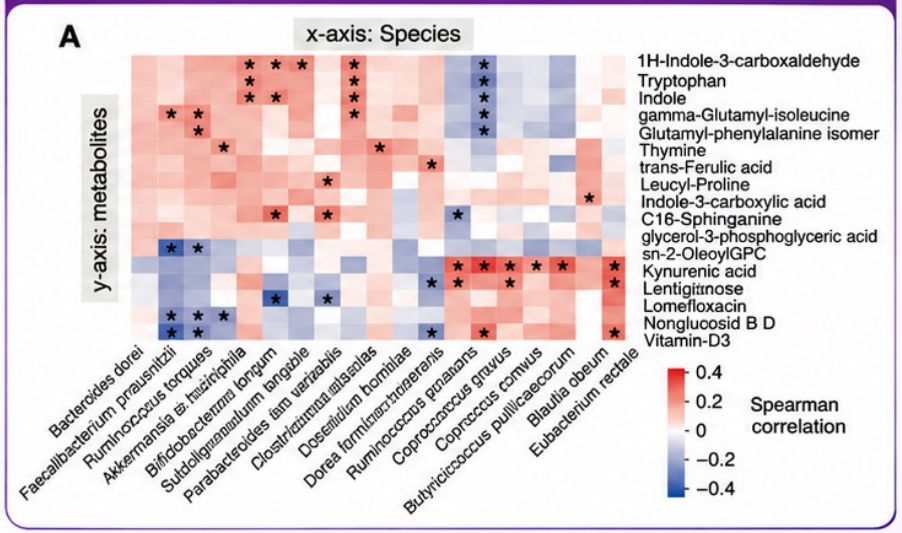
Each molecular layer measures a different part of biology.

Microbiome Patterns in Context

Alpha diversity and beta diversity



Microbe–metabolite correlations



Compositional data

Covariates matter

Association ≠ cause

Oh et al., Cell Metabolism (2020), Fig. 1B–C and Fig. 4A. Original data panels.

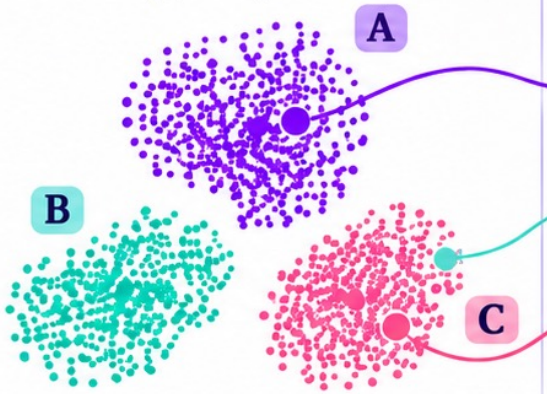
Microbiome associations require careful sampling context.

Cell Identity, State, and Location

Illustrative schematic: three distinct biological questions

Cell identity

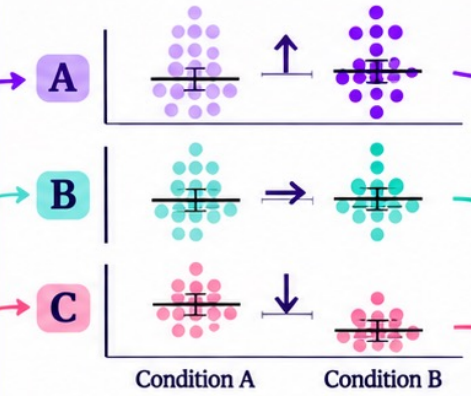
scRNA-seq (UMAP)



Which population?

Cell state

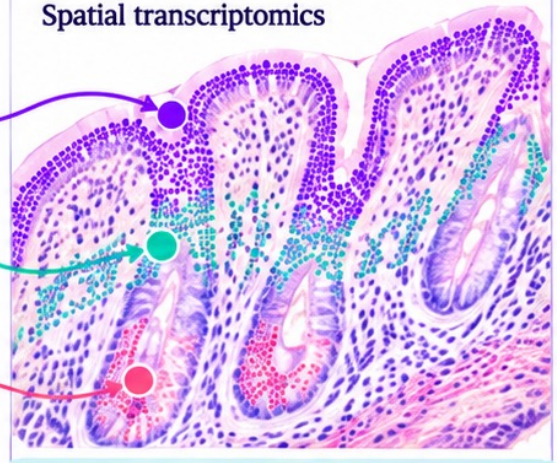
Gene expression (per donor)



Within-cell-type change?

Spatial context

Spatial transcriptomics



Where in tissue?

Donors are biological replicates

- Donor 1
- Donor 2
- Donor 3
- ...

Illustrative teaching schematic; cell clusters and spatial signals are not measured study data.

Cell-level patterns require donor and tissue context.

What Does Integration Add?

Paired host transcriptome + microbiome

Colorectal cancer
CRC (n = 44) | Non-CRC (n = 44)

Inflammatory bowel disease
IBD (n = 56) | Non-IBD (n = 22)

Irritable bowel syndrome
IBS (n = 29) | Non-IBS (n = 13)

Colonic mucosal biopsy

Gut microbiome → 16S rRNA sequencing → Gut microbiome abundance (X)

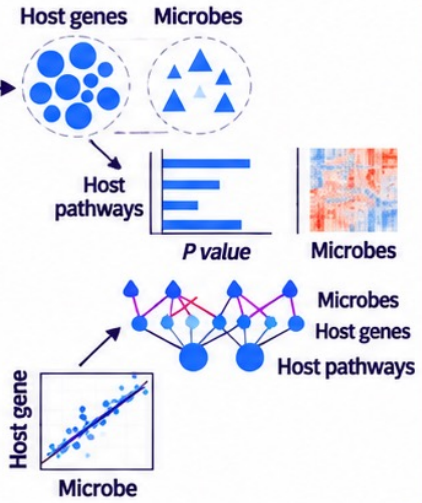
Host mRNA → RNA sequencing → Host gene expression (Y)

Sparse CCA
$cor(X_u, Y_v) + \lambda_{1u} + \lambda_{2v}$

- Hyperparameter tuning
- Fit sparse CCA model (LOOCV)
- Multiple testing correction (FDR)
- Enrichment analysis for host gene sets

Lasso
$y = \beta X$

- Lasso model fitting (LOOCV)
- Regularized projection
- Stability selection
- Multiple testing correction (FDR)



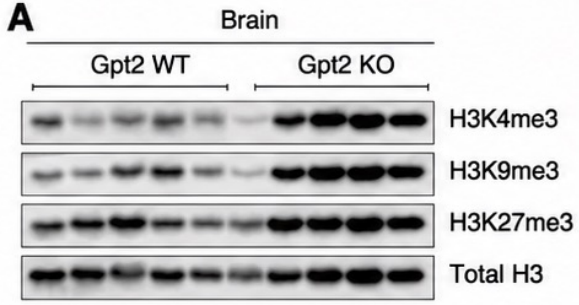
Matched samples → **Association analysis** → **Pathway hypothesis**

Priya et al., Nature Microbiology (2022), Fig. 1. Original study-integration workflow.

Integration connects assays without proving causality.

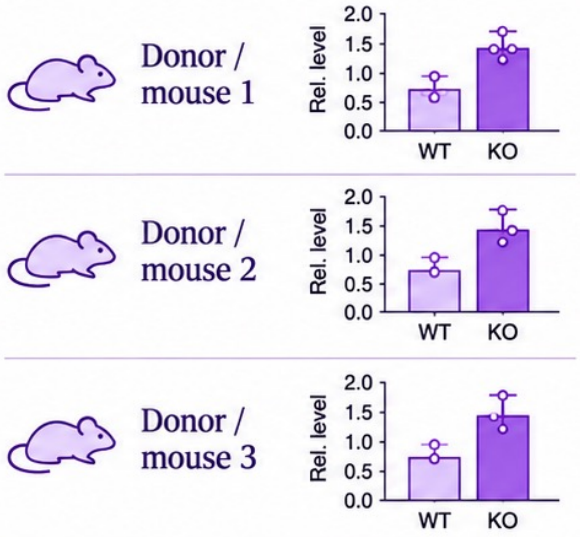
What Does Validation Establish?

Orthogonal measurement

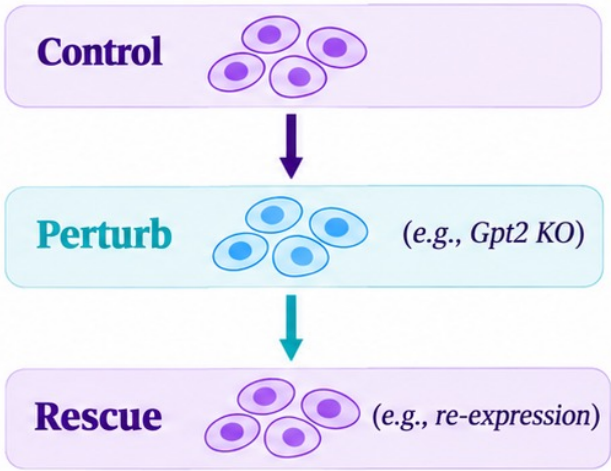


Histone-mark immunoblot

Independent replication



Perturbation and rescue



Original assay: Sternisha et al., Science (2026), Fig. 6A. Replication/rescue diagrams are instructional.

Validation should target the unresolved biological question.

Which Claim Does the Model Support?

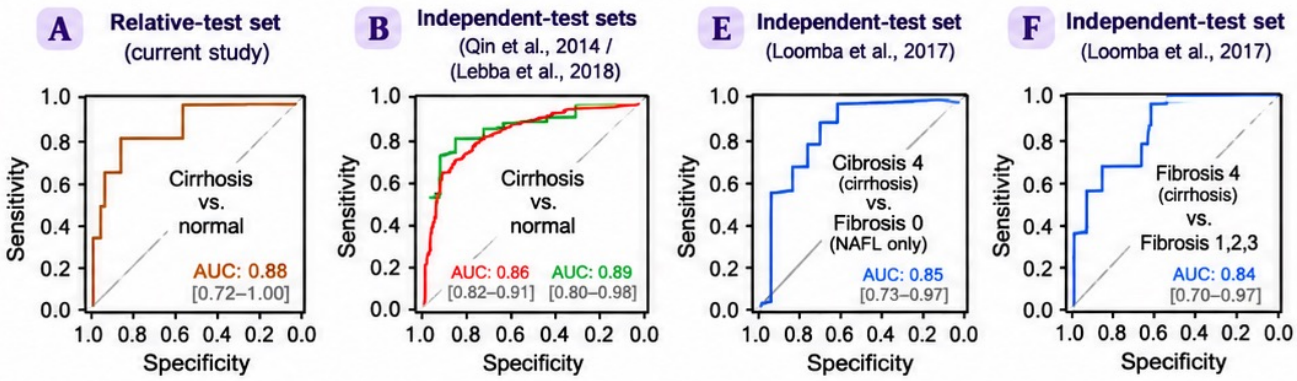
External validation: ROC curves

Validation in relatives and external cohorts

Model: [19 species + age]

Validation in mixed fibrosis cohort

Model: [19 species + age]



Which statement fits?

- Causes cirrhosis
- Discriminates groups
- Ready for clinical use

Oh et al., Cell Metabolism (2020), Fig. 5A–B, E–F. Original curves; question is a class exercise.

Predictive discrimination does not establish a causal mechanism.

The Structure of a Scientific Paper

ABSTRACT: the complete story in brief

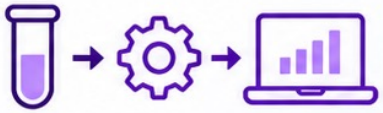
INTRODUCTION



- Research background
- Knowledge gap
- Study objective / hypothesis

Why this question?

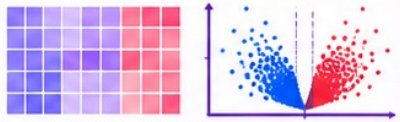
METHODS



- Data / samples
- Experimental procedures
- Computational analysis
- Key parameters (reproducibility)

How was it tested?

RESULTS



- Main findings (with evidence)
- Key comparisons
- Statistical support
- Figure / table summary

What was found?

DISCUSSION



- Biological meaning
- Comparison with prior work
- Limitations and alternatives
- Implications and future work

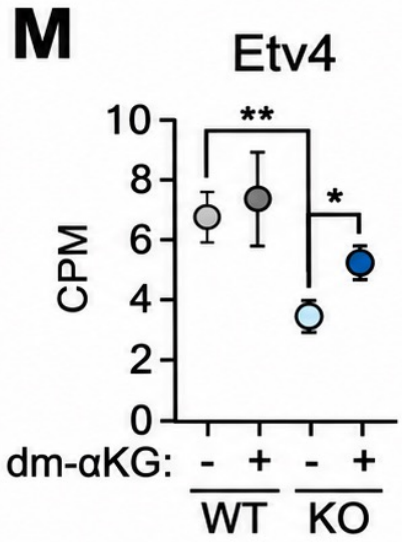
What does it mean?

Teaching framework for an IMRaD-style research article; actual journal structures may vary.

Each section answers a different reader question.

The Facts a Results Paragraph Needs

Original evidence: Etv4



Six facts to record

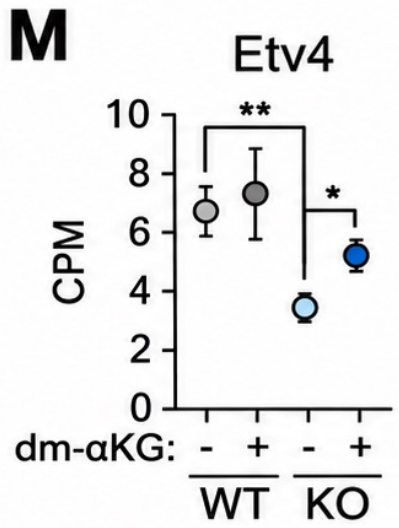
Comparison	KO + dm-αKG vs KO vehicle
Direction	Higher Etv4 RNA abundance
Measurement	RNA-seq CPM
Biological n	Vehicle 10; treated 8
Summary	Group means ± SEM
Uncertainty	Exact estimate / CI: check source

Sternisha et al., Science (2026), Fig. 6M. RNA-seq CPM, group means ± SEM.

Extract the measured facts before writing the claim.

From a Figure to Two Results Sentences

Read the figure



Write what the evidence supports

OBSERVATION

In Gpt2-knockout mouse brain, dm-αKG treatment increased Etv4 RNA abundance relative to vehicle (Fig. 6M; $P < 0.05$).

INTERPRETATION / LIMIT

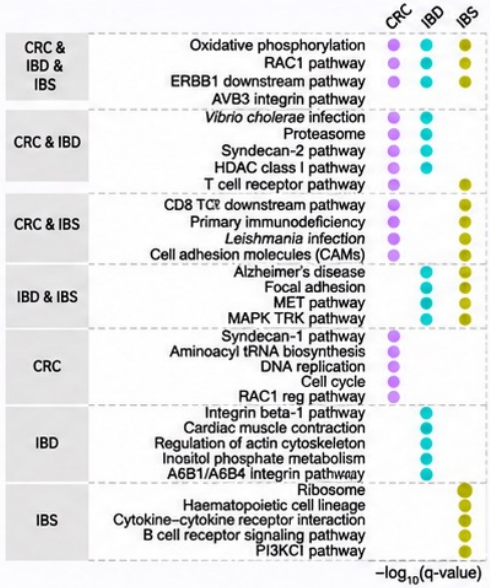
The finding is consistent with an αKG-responsive expression change, but it does not alone establish direct chromatin-mediated control.

Teaching paraphrase based on Sternisha et al., Science (2026), Fig. 6M; verify exact statistics in original legend.

A strong Results sentence states both the result and its scope.

Your Turn: A Claim That Fits the Data

READ THE EVIDENCE



Host pathways associated with gut microbes

REWRITE THE CLAIM

OVERSTATED CLAIM

"These microbes drive the host disease pathway."

YOUR TASK — 3 MINUTES

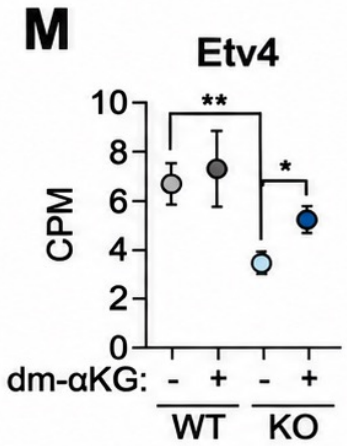
- Write one observation supported by the panel.
- Write one interpretation that preserves uncertainty.
- Name the experiment needed for a stronger claim.

Real study figure: Priya et al., Nature Microbiology (2022), Fig. 2a.

Use verbs that match the strength of the available evidence.

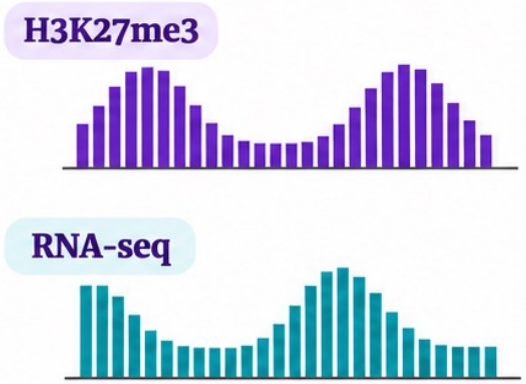
The Logic Between Results

INITIAL FINDING



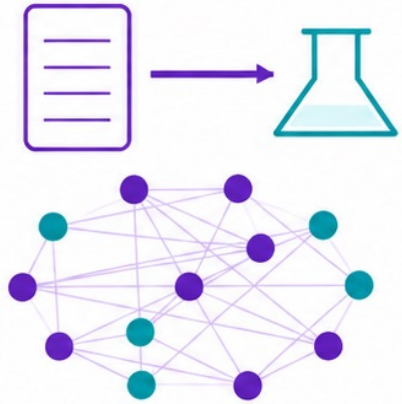
Observe an expression difference

UNRESOLVED QUESTION



Could regulation explain the change?

NEXT EXPERIMENT



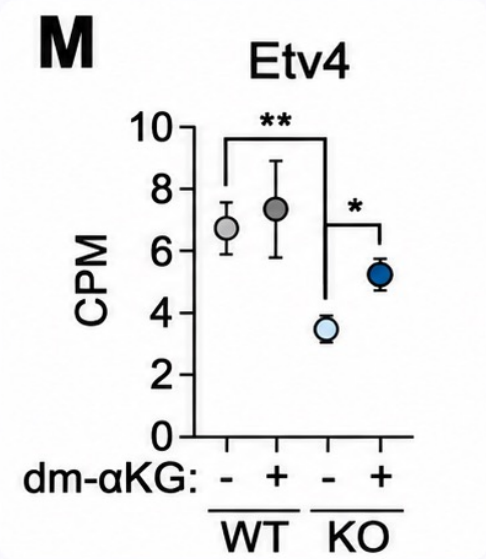
Test the candidate mechanism

Illustrative question-driven sequence based on the reading and writing framework.

Each Results section should explain why the next experiment was needed.

What a Figure Legend Must Explain

FIGURE 6M: ETV4



THE SELF-CONTAINED LEGEND

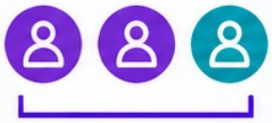
- What **was measured?** (Etv4 RNA abundance)
- **Which biological groups** and treatments?
- **How many** independent biological units?
- What do points and **error bars** represent?
- Which **test and comparisons** were used?
- What do the **significance labels** mean?

Example figure: Sternisha et al., Science (2026), Fig. 6M; inspect the original legend.

A figure legend supplies the information needed to read the figure.

Methods That Support Reproduction

SAMPLES & DESIGN



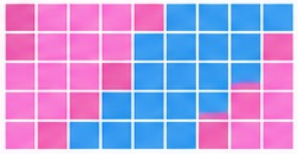
- Biological n
- Groups and exclusions
- Pairing, batch, covariates

PREPROCESSING



- Data and reference
- QC and filtering
- Normalization / missingness

STATISTICAL MODEL



- Design and contrasts
- Statistical model
- Multiplicity control (FDR)

CODE & PROVENANCE



- Tools and versions
- Parameters and scripts
- Data accessions

Reproducible Methods report the decisions that can change the result.

The Gap That Motivates a Study

FROM BACKGROUND TO OBJECTIVE

BIOLOGICAL PROBLEM
What is the broad issue?

KNOWLEDGE GAP
What remains uncertain?

STUDY OBJECTIVE
What will be tested?

PUBLISHED CASE EXAMPLE

Problem
Host–microbe relationships in gastrointestinal disease

Gap
Shared versus disease-specific pathways remain uncertain

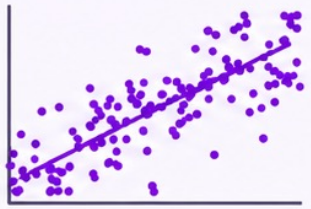
Objective
Integrate paired host expression and microbial profiles

Teaching example adapted from Priya et al., Nature Microbiology (2022).

The Introduction makes the study question necessary and understandable.

Meaning, Alternatives, and Limits

RESULT



Association

MEANING
Candidate relationship

ALTERNATIVE
Shared confounding

LIMIT
Observational context

NEXT TEST

Independent replication or a targeted perturbation

Association schematic for teaching; not a quantitative result from the cited studies.

A useful Discussion explains both the contribution and its boundaries.

An Abstract That Matches the Evidence

FOUR QUESTIONS

01 QUESTION
Why investigate?

02 APPROACH
How was it studied?

03 FINDING
What was observed?

04 IMPLICATION
What follows?

EXAMPLE — EVIDENCE-MATCHED ABSTRACT

Host gene expression and gut microbial composition may be related in gastrointestinal disease, but shared and disease-specific associations are incompletely understood. The authors integrated paired host expression and microbial profiles across disease cohorts. They identified both shared and disease-specific gene–microbe associations and enriched host pathways. These findings prioritize candidate relationships for experimental investigation; the observed associations do not by themselves establish direct microbial effects.

Example is a teaching paraphrase of Priya et al., Nature Microbiology (2022), not a direct quotation.

The title and abstract must preserve the scope of the evidence.

Sources and AI-Assisted Writing

DISTINGUISH THE SOURCE

PUBLISHED STUDY
Claim made in the paper

YOUR REANALYSIS
Result from your code

YOUR HYPOTHESIS
Interpretation to be tested

EVIDENCE RECORD
Figure / dataset / n

→

DRAFT
Human or AI-assisted

→

VERIFY
Recheck sources

Every number, citation, and scientific claim needs a verifiable source.

Your Week 12 Writing Portfolio



EVIDENCE MAP

Claim ↔ panel



RESULTS

Two linked paragraphs

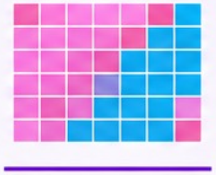


FIGURE LEGEND

Self-contained caption



DISCUSSION

Meaning and limits



METHODS RECORD

Reproducible decisions

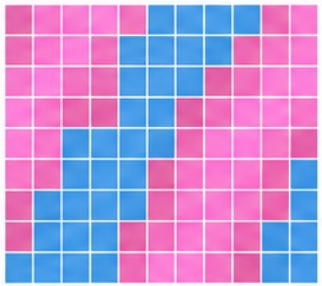
- Use at least two omics layers in your scientific argument.
- Trace every major statement to a figure, result, or source.



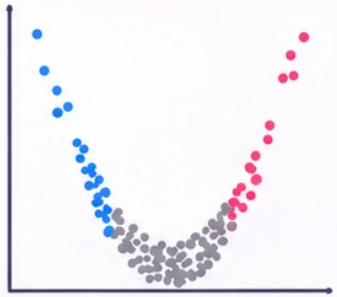
Your portfolio should connect each written claim to its source evidence.

From Evidence to a Scientific Claim

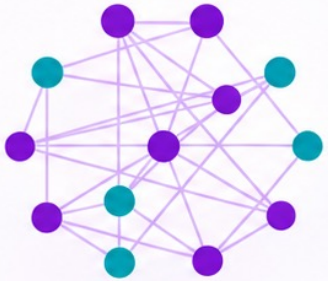
EVIDENCE



OMIC DATA



STATISTICAL RESULTS



INTEGRATED EVIDENCE

YOUR SCIENTIFIC CLAIM

What can you conclude?
State the supported result.

What remains unresolved?
Name the next test.

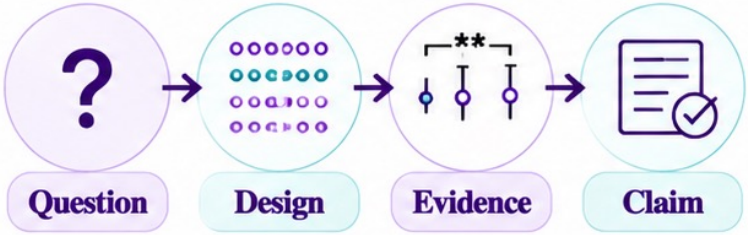
Write the strongest conclusion that the evidence can support.

Summary

KEY TAKEAWAYS

- Claims are testable scientific questions.
- Evidence links design to a supported answer.
- Interpretation must account for effect size, uncertainty, and FDR.

FROM QUESTION TO ANSWER



Multi-omics data → Biological discovery

PRACTICAL PRINCIPLES

- Match methods to the question.
- Use appropriate controls and comparisons.
- Report and interpret results with effect size, uncertainty, and FDR.

A principled, evidence-based workflow turns multi-omics data into reliable discovery.